

BRENDAN HARRIS

Postdoctoral Research Associate (2026–)

Complex Systems Group, School of Physics
The University of Sydney

✉ brendan.harris@sydney.edu.au
🆔 0000-0003-3412-4186
🌐 [Personal website](#)
📍 Sydney, Australia

🐙 [Github](#)
🏠 [Google Scholar](#)
🦋 [Bluesky](#)
☎ +61 466 956 165

RESEARCH THEMES

I study the principles of neural computation, developing new analytical methods to connect noisy, high-dimensional recordings to theory from statistical physics and dynamical systems:

- **Spatiotemporal dynamics:** how nested traveling waves coordinate hierarchical neural processing and flexible behavior.
- **Criticality:** robust methods to track the distance to criticality in noisy, poorly sampled systems.
- **Stochastic processes:** fractional diffusion for neural mean field, decision-making, and visual sampling.
- **Population geometry:** low-dimensional coding, communication subspaces, and dimensionality reduction.

PUBLICATION RECORD

6 peer-reviewed publications, 73 citations, h-index 5. 2 first-author papers in Q1 journals (*Physical Review X*, top 3% General Physics and Astronomy, Scopus CiteScore; *Nature Communications*, top 3% Multidisciplinary). All first-author work accompanied by [open-source code](#).

CONTENTS

Education	Workshops and talks	Industry
Publications and Preprints	Supervision	Awards
Conferences	Teaching	Referees

EDUCATION

2022 – 2026	Physics PhD <i>Cross-scale dynamics in the working regime of the visual cortex</i>	The University of Sydney
2021	Physics Honours, Class I and the University Medal <i>Inferring parametric variation across non-stationary time series</i>	The University of Sydney
2018 – 2020	Bachelor of Science/Bachelor of Advanced Studies (Dalyell Scholar) Majored in Physics and Neuroscience	The University of Sydney

PUBLICATIONS AND PREPRINTS

2026	Anomalous dynamics in the working regime of the visual cortex Brendan Harris and Pulin Gong	In preparation
2026	Nested spatiotemporal theta–gamma waves organize hierarchical processing across the mouse visual cortex 🏠🔗 Brendan Harris and Pulin Gong	Nature Communications vol. 17, pp. 2629
Jul 2025	Canonical time-series features for characterizing biologically informative dynamical patterns in fMRI 🏠🔗 Imran Alam, Brendan Harris , Patrick Cahill, Oliver Cliff, Marija Markicevic, Valerio Zerbi, and Ben D. Fulcher	Aperture Neuro vol. 5, pp. 140433
Aug 2024	Tracking the Distance to Criticality in Systems with Unknown Noise 🏠🔗 Brendan Harris, Leonardo L. Gollo, and Ben D. Fulcher	Physical Review X vol. 14, pp. 031021

May 2024	<i>Distributed and dynamical communication: a mechanism for flexible cortico-cortical interactions and its functional roles in visual attention</i>  Shencong Ni, Brendan Harris , and Pulin Gong	Communications Biology vol. 7, pp. 550
Jul 2021	<i>Approximate Modal Cut-Off Wavelengths and the V-Parameter for M-type Optical Fibers and Its Novel Applications</i> Deepak Jain, Mark A George, Brendan Harris , and Simon Fleming	Journal of Lightwave Technology vol. 39, pp. 4478–4488
CONFERENCES		
Jun 2025	OHBM 2025  Brisbane, QLD, Australia <i>2072: Nested spatiotemporal dynamics organize hierarchical processing in the mouse visual cortex</i> Brendan Harris and Pulin Gong	Poster
Jun 2025	EPC/APCV 2025 The University of New South Wales <i>Spatiotemporal theta–gamma waves organize hierarchical processing in the mouse visual cortex</i> Brendan Harris and Pulin Gong	Talk
Mar 2025	COSYNE 2025  Montreal, QC, Canada <i>1–117: Tracking the distance to criticality across the mouse visual hierarchy</i> Brendan Harris , Leonardo Gollo, and Ben Fulcher	Poster
Feb 2025	NeuroEng 2025 The University of Melbourne <i>Nested spatiotemporal theta–gamma waves organize hierarchical visual processing</i> Brendan Harris and Pulin Gong	Talk
Jul 2024	2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) <i>1–4: Default Mode Network Detection using EEG in Real-time</i> Navin Cooray, Chetan Gohil, Brendan Harris , Shaun Frost, and Cameron Higgins	Conference paper
Oct 2023	IBRO Neuroscience Reports (IBRO World Congress 2023) Granada, Spain <i>S783: Nested and non-stationary oscillatory bursts underlie flexible inter-areal neural communication</i> Brendan Harris and Pulin Gong	Poster
Jul 2022	31st Annual Computational Neuroscience Meeting: CNS*2022 Melbourne, Australia <i>P36: Summarizing non-stationarity in spatio-temporal neural data</i> Brendan Harris and Ben Fulcher	Poster
Jun 2022	28th Annual Meeting of the Organization for Human Brain Mapping <i>Summarizing non-stationarity in spatio-temporal neural data</i> Brendan Harris and Ben Fulcher	Virtual Poster
Sep 2020	Frontiers in Optics / Laser Science (2020) <i>FM4D.2: Approximate normalized frequency (V-parameter) and modal cut-off wavelengths of M-type optical fibers</i> Deepak Jain, Mark George, Brendan Harris , and Simon Fleming	Oral presentation

WORKSHOPS AND TALKS

Jul 2026	Sydney – Tokyo: Neural and Nonlinear Dynamics Workshop <i>Beyond PCA: Isolating dataset-specific constraints from high-dimensional data</i>	Talk
Jun 2026	Neural Systems, Behaviour Lab, Turner Institute for Brain, and Mental Health, Monash University <i>Nested spatiotemporal θ-γ waves organize hierarchical visual processing</i> Invited to give a talk on my PhD research	Talk
Apr 2026	Maths in the Brain  <i>Anomalous dynamics in the working regime of the visual cortex</i> Prize: “Best ECR talk”	Talk
May 2025	Center for Neuroscience Imaging Research, IBS, Republic of Korea <i>Nested spatiotemporal θ-γ waves organize hierarchical visual processing</i> Invited to give a talk on my PhD research	Talk
Mar 2025	Pre-COSYNE BrainHack 2025 Montreal, QC, Canada Workshop on open science and open-source tools for computational and systems neuroscience.	Workshop attendee
Aug 2024	Strong Compute Chess AI Hackathon Sydney, NSW, Australia Hackathon on transformer models and distributed GPU training for chess	Workshop attendee
Aug 2024	Emerging Aspirations in Complex Systems The University of Sydney <i>Spatiotemporal θ-γ waves in the mouse visual cortex</i> Centre for Complex Systems, The University of Sydney	Workshop talk
May 2024	School of Physics HDR symposium The University of Sydney <i>Spatiotemporal θ-γ waves in the mouse visual cortex</i> School of Physics, The University of Sydney	Workshop talk
Oct 2023	MIP:Lab, Geneva Biotech Campus Geneva, Switzerland <i>Burst-based inter-areal neural communication</i> Presented my PhD work to the MIP:Lab, hosted by Dr. Enrico Amico.	Talk
Jul 2023	Cognitive Neuroscience Hub, The University of Melbourne The University of Melbourne <i>Burst-based inter-areal neural communication</i> Presented my PhD work, hosted by Dr. Jacob Paul.	Talk
Jul 2023	tLab Monash University <i>Burst-based inter-areal neural communication</i> Presented my PhD work to the lab of Professor Nao Tsuchiya.	Talk
Jan 2023	Complexity, Criticality and Computation Symposium: C3-2023 Heron Island <i>Burst-based inter-areal neural communication</i> Symposium held by the Centre for Complex Systems (The University of Sydney).	Workshop talk
Nov 2022	BrainHack Australia 2022 The University of Sydney <i>Brainharmonic</i> Contributed to a project aiming to synthesize music from neural signals using deep neural networks.	Workshop attendee

Oct 2022	OpenScope NeuroDataReHack Hackathon Seattle, WA, USA Workshop on re-analyzing open-source neural data, hosted by the Allen Institute	Workshop attendee
Oct 2022	Yale Medical School <i>Feature-based analysis of neural time series</i> Invited by Dr. Evelyn Lake to present the results of my projects involving time-series features to a journal club.	Talk
Aug 2022	Emerging Aspirations in Complex Systems The University of Sydney <i>Inferring parametric variation across non-stationary neural time series</i>	Workshop talk
Jul 2022	31st Annual Computational Neuroscience Meeting: CNS*2022 Melbourne, Australia <i>Inferring parametric variation across non-stationary neural time series</i> In workshop 'Highly comparative analysis of neural dynamics'.	Workshop talk
Dec 2021	Neuromatch 4.0  <i>Summarizing non-stationarity in spatio-temporal neural data</i>	Flash talk

SUPERVISION

Aug – Nov 2026	Physics 3rd-Year Interdisciplinary Project <i>The University of Sydney</i> Bunyodjon Sattorov, Murray Jones, Noah Scollard, Shelly Chen, Yiying Lim	Secondary
Aug – Nov 2026	Physics 3rd-Year Interdisciplinary Project <i>The University of Sydney</i> Andrew Booth, Huu Hoang Guong Nguyen, Isaac Palazzi, Kiran McLernon	Primary
Mar – Nov 2026	Physics Honours student <i>The University of Sydney</i> Jude Metcalf “Adaptive fractional dynamics of visual sampling”	Secondary
Mar – Nov 2025	Physics Honours student <i>The University of Sydney</i> Aiden Sloots “Bi-fractional diffusion for decision-making dynamics”	Secondary

TEACHING

Jul – Nov 2024	Computational optics <i>The University of Sydney</i> Tutored a computational optics lab	Tutor
Feb – Mar 2023	3rd-year Statistical Mechanics <i>The University of Sydney</i> Tutored a combined lecture and tutorial class on statistical mechanics, covering topics such as the Einstein solids, the Ising model, Peierls argument, Monte-Carlo sampling, phase transitions, criticality, and the renormalization group.	Tutor

Aug 2022 – Mar 2023	3rd-year Interdisciplinary Physics computational lab <i>The University of Sydney</i> Tutored the Matlab computational labs of PHYS3888, a course on the physics of complex systems that introduces third year physics students to networks, linear and nonlinear dynamical systems, criticality, and machine learning/neural networks.	Tutor
Mar – Nov 2022	Second-year physics Matlab course <i>The University of Sydney</i>	Tutor
Aug – Nov 2018	Science Dalryell Showcase <i>The University of Sydney</i>	Mentor

INDUSTRY

Mar – Dec 2023	Resonait Medical Technologies <i>Dr. Cameron Higgins</i> Worked on a casual basis writing Matlab code to interface with EEG headsets, infer the activation of brain networks, and implement neurofeedback for the monitoring and treatment of depression	Software developer
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AWARDS

2026	Best ECR talk prize <i>Turner Institute for Brain and Mental Health</i> Maths in the Brain 2026	Prize
2025	Postgraduate Research Support Scheme <i>The University of Sydney</i> To fund travel and accommodation for the COSYNE 2025 conference (Montreal, Canada)	Travel grant
2025	COSYNE Presenters Travel Grant <i>COSYNE</i> To fund travel and accommodation for the COSYNE 2025 conference (Montreal, Canada)	Travel grant
2023	Postgraduate Research Support Scheme <i>The University of Sydney</i> To fund travel and accommodation for the IBRO2023 conference (Granada, Spain)	Travel grant
2022	Dean's List of Excellence in Academic Performance <i>The University of Sydney</i> For academic performance during my Honours year	Recognition
2022	Honours Class I and the University Medal <i>The University of Sydney</i> For academic and research performance during my Honours year	Recognition
2022	Research Training Program (fee offset and stipend) <i>The University of Sydney</i> To fund my PhD studies	Scholarship
2022	Emerging Aspirations in Complex Systems Award <i>The University of Sydney</i> To fund accommodation for the C3 symposium	Travel grant

2021	The University of Sydney Physics Foundation Scholarship No III <i>The University of Sydney</i> For academic performance in 3rd year undergraduate Physics	Scholarship
2021	University of Sydney Academic Merit Prize <i>The University of Sydney</i> For academic performance during my Honours year	Prize
2019	Denison Research Scholarship and Denison Relocation Scholarship <i>The University of Sydney</i> To fund a 6-week summer research project with the Quantum Control Laboratory	Scholarship
2018	Dean's List of Excellence in Academic Performance <i>The University of Sydney</i> For academic performance in my 1st year of undergraduate study	Recognition
2017	Sydney Scholars Award <i>The University of Sydney</i> For academic results in the higher school certificate	Scholarship

REFEREES

A/Prof. Benjamin D. Fulcher

School of Physics
The University of Sydney

Physics Building, Physics Rd
Camperdown 2006

Ph: +61 481 563 731
ben.fulcher@sydney.edu.au

Prof. Pulin Gong

School of Physics
The University of Sydney

Physics Building, Physics Rd
Camperdown 2006

Ph: +61 2 9036 9368
pulin.gong@sydney.edu.au

Dr. Cameron Higgins

Resonait Medical Technologies

Terrigal, NSW

Ph: +61 423 344 596
cam@resonait.com